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3 January 1950

MILITARY SPECIFICATION

FITTINGS; FLUID CONNECTION

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 Fittings shall be furnished in the types and styles designated by the applicable AN Standard Detail drawings most of which are referred to on Drawing AND10059.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified::

2.1.1 Specifications.-

Federal

QQ-A-354	Aluminum-Alloy (AL-24)(Aluminum-Copper-Magnesium (1.5%)-Manganese); Bars, Rods, Shapes, and Wire
QQ-A-355	Aluminum-Alloy (AL-24)(Aluminum-Copper-Magnesium (1.5%)-Manganese); Plates, Sheets, and Strips
QQ-A-367	Aluminum-Alloy; Forgings, Heat-Treated
QQ-B-611	Brass, Commercial; Bars, Plates, Rods, Shapes, Sheets, and Strips
QQ-B-721	Bronze, Manganese; Bars, Forgings, Plates, Rods, and Shapes, Sheets, and Strips
QQ-M-151	Metals; General Specification for Inspection of
QQ-P-416	Plating; Cadmium (Electrodeposited)
QQ-S-624	Steel; Alloy; Bars (General Purpose)
QQ-S-633	Steel, Carbon; Bars (General Purpose)

Military

MIL-B-6946	Bronze, Aluminum; Bar and Rod
MIL-H-5440	Hydraulic Systems; Design, Installation and Tests of Aircraft (General Specification for)
MIL-H-6088	Heat Treatment of Aluminum Alloys; Process for (For Aircraft Applications)
MIL-H-6875	Heat Treatment of Steels (Aircraft); Process for
MIL-S-6050	Steel; Chrome-Nickel-Molybdenum (8630) Bars, Rods, and Forging Stock (For Aircraft Applications)
MIL-S-6049	Steel; Chrome-Nickel-Molybdenum (8740) Bars, Rods, and Forging Stock (For Aircraft Applications)
MIL-S-6758	Steel; Chrome-Molybdenum (4130) Bar, Rod, and Forging Stock (For Aircraft Applications)
MIL-T-6732	Tubing; Chrome-Nickel-Molybdenum (8630) Steel Seamless

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JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes; Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes; Fiberboard (V-Board and W-Board) Exterior and Interior
JAN-P-120	Packaging and Packing for Overseas Shipment - Cartons; Folding, Paperboard
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier-Materials; Waterproof, Flexible
JAN-P-133	Packaging and Packing for Overseas Shipment - Boxes; Set-Up, Paperboard
JAN-P-139	Packaging and Packing for Overseas Shipment - Plywood; Container Grade

Air Force-Navy Aeronautical

AN-QQ-A-696	Anodic-Films; Corrosion-Protective (for) Aluminum Alloys
AN-P-13	Preservation and Packaging; Parts and Equipment (General Specification for)
AN-P-363	Pipe-Threads; Taper, Aeronautical National Form, Symbol ANPT
AN-S-9	Steel; Molybdenum (4037) Bar and Rod
AN-S-126	Screw-Threads; Standard, Aircraft

U. S. Army

94-40645	Markings; Exterior, Domestic, and Export Shipment, by Contractors

2.1.2 Other Publications.-

Bureau of Supplies and Accounts

Navy Shipment Marking Handbook

2.1.3 Drawings.-

Air Force-Navy Aeronautical Standard Drawings

AN819	Sleeve - Coupling
AND10050	Bosses - Standard Dimensions for Gasket Seal Straight Thread
AND10056	Fitting End - Standard Dimensions for Flared Tube Connection and Gasket Seal
AND10057	Fitting End - Standard Dimensions for Bulkhead Flared Tube Connection
AND10058	Fitting End - Hose Connection (Standard Dimensions For)
AND10059	Fittings - Styles of Pipe Thread - Flared Tube and Hose
AND10064	Fittings - Assembly of Universal and Bulkhead Flared Tube

(Copies of this specification and copies of other publications referenced herein or required for Government procurement, and the Index of Military Aeronautical (AN or MIL) Standards, may be obtained upon application to the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio; or the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania.)

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3. REQUIREMENTS

3.1 Material.- Fittings shall be fabricated of materials listed in table I as specified on the applicable AN drawing.

3.1.1 Steel.- All steel shall be made by the open hearth or electric furnace process.

3.1.2 Heat Treatment.-

3.1.2.1 Aluminum Alloy.- Aluminum alloy materials shall be heat treated in accordance with Specification MIL-H-6088.

3.1.2.2 When required, steels shall be heat treated in accordance with Specification MIL-H-6875.

3.1.2.3 Fittings made of aluminum-bronze, conforming to Specification MIL-B-6946, shall be heated to and held at 1050 ±10°F for 1-1/2 hours to 2 hours and then allowed to cool in still air. This heat treatment shall be accomplished after machining and before plating.

3.1.3 Internal Strain of Aluminum-Bronze and Copper-Base Alloys.- Fittings made of aluminum-bronze and copper-base alloys shall have no internal strain as revealed by the test specified in Section 4.

3.1.4 Physical Properties.- Unless otherwise specified on the applicable AN detail drawings the hardness of the finished steel parts shall be Rockwell C10-31. Physical properties of copper-base alloy forgings shall be governed by the physical properties of the as-received material as listed in table I. For steel fittings falling within the hardness range of Rockwell C10-20, hardness tests shall be made using the Rockwell B scale, hardness values to fall in the range of Rockwell B92-99.

TABLE I
Materials

Material		Specification Number	
Aluminum Alloy	Bars, Shapes, Sheet, Strip and Plate	QQ-A-354	Cond T
		QQ-A-355	Cond T
	Forgings	QQ-A-367 Class 5	Cond T6
Steel	Bars, Rods, and Forgings	MIL-S-6758	
		AN-S-9	
		MIL-S-6050	
		QQ-S-633	FS1117
			FS1118
			FS1137
		QQ-S-624	FS1335
		MIL-S-6049	
	Tubing; Seamless	MIL-T-6732	
Aluminum-Bronze	Bar and Rod	MIL-B-6946	
Copper-Base Alloys	Bar, Rod, Shapes and Forgings	QQ-B-721 Half Hard 1/	
	Bar and Rod	QQ-B-611 Comp B	
	Forgings	QQ-B-611 Comp A	

1/ With bronze-forgings a soft temper will be permitted in the finished part.

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3.1.5 Alternate Material.- Materials not listed in table I and use of which has not been specifically prohibited may be used only upon specific approval by the Air Force and Navy. In order to obtain this approval it will be necessary to conduct complete Performance tests for alternate materials and submit a test report as specified herein. In no case shall the alternate material be used in production fittings prior to receipt of specific approval from the Services.

3.1.6 Except when specifically allowed on the AN Standard detail drawing, castings of any type shall not be used.

3.1.7 Fittings made by welding or brazing together two or more separate pieces of metal may be used on service aircraft only when the design method of fabrication by each manufacturer offering such fittings has been submitted to and approved by the Air Materiel Command and the Bureau of Aeronautics.

3.2 Design and Dimensions.- The design and dimensions of AN Standard fittings shall be in accordance with the applicable AN Standard detail drawings. The bosses, and tube and hose ends of fittings shall be in accordance with Drawings AND10050, AND10056, AND10057, or AND10058, as applicable.

3.2.1 Shape and Form Restrictions.- Abrupt reductions of section shall be avoided. Small external sections adjoining relatively heavy body sections shall be shaded into the heavier sections by means of ample fillets. Sharp corners or inadequate fillets, excessive undercuts or excessive grooves at the junction of such small sections with large sections of fittings shall be considered cause for rejection. Unless otherwise specified, drawing tolerances shall apply to fillet and corner radii.

3.2.2 Passages.-

3.2.2.1 Drill Offset.- On straight fittings where the fluid passage is drilled from each end, the offset between the drilled holes at the meeting point of the drills shall not exceed 0.015 inch. It shall be possible to pass through the fluid passage a ball whose minimum diameter is 0.020 less than the minimum diameter specified for the passage. This does not mean that the drilled passage may be smaller than that required by the detail drawings.

3.2.2.2 On angle fittings, the cross-sectional area at the junction of the fluid passages shall be not smaller than the cross-sectional area of the smaller passage.

3.2.2.3 Wall Thickness.- Except as otherwise specified on the applicable AN drawing, the wall thickness at any point on the fitting shall be not less than the thickness established by the dimensions and tolerances for the inside and outside diameters and eccentricities specified in the AN detail drawing.

3.2.2.4 Threads.- Except as otherwise specified, all straight threads shall conform to Specification AN-S-126. Pipe threads shall conform to Specification AN-P-363.

3.3 Finish.- All fittings shall be finished as noted in table II in accordance with the applicable finish specification.

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TABLE II
Finish

Fitting Material	Type of Finish	Finished Specification
Aluminum-Alloy	Anodic Treatment	AN-QQ-A-696
Steel	Zinc Plate or Cadmium Plate	AN-P-32 (Black Colored) or QQ-P-416, Type II, Class C (Black Colored)
Aluminum Bronze	Cadmium Plate	QQ-P-416 Type I, Class C
Copper-Base Alloys	Cadmium Plate	QQ-P-416 Type I, Class C

3.4 Identification of Product.- Unless otherwise specified all fittings shall be permanently marked with the letters "AN" and the manufacturer's name or trademark. If impression stamping is not used the method of marking shall be subject to the approval of the Air Force and Navy. The letters "AN" shall be separate and distinctive from the manufacturer's name or trademark. This marking shall be applied in a location not detrimental to the fitting and shall not be detrimental to the corrosion protection of the fitting. In addition, AN Standard fittings shall be identified by the following coloring:

Aluminum-Alloy	-	Light Blue
Steel	-	Black
Aluminum-Bronze	-	Natural Cadmium Plate
Copper-Base Alloys	-	Natural Cadmium Plate

Where application of color is necessary it shall be applied by dyeing, or other process, to provide a color which will not rub off or be smeared by contact incident to handling and service and which will not injure the material, clog the threads, or interfere in any way with the functioning of the fittings in service. If the coloring material used is soluble or would affect in any way the liquids used for fuel, lubricant, coolant, hydraulic, anti-icing or oxygen systems of airplanes, those areas of the fittings coming in contact with such fluids shall not be colored. The dye shall be capable of withstanding immersion in a mild alkaline cleaning solution approved for cleaning ferrous and nonferrous alloys.

3.5 Weight.- The maximum weight of fittings shall not exceed the values on the applicable AN drawings.

3.6 Workmanship.- Fittings shall be free from cracks, laps, seams, burrs, longitudinal and spiral tool marks, or any other defects which may detrimentally affect their suitability for the service intended. All sealing surfaces shall be smooth except that annular tool marks up to 100 microinches RMS maximum will be acceptable. Contractor's production and inspection processes shall be established so the end results will produce a completed article in strict compliance with specification and contract requirements.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.- All the tests required for the testing of fittings are classified as Inspection tests, for which necessary sampling techniques and methods of testing are specified in this section. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities

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satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the Procuring Service. The contractor shall maintain a record, showing quantitative results for all tests required by this specification, and available, to the Inspector, signed by an authorized representative of the contractor or laboratory, as applicable. Certificates showing conformance with the applicable material specification shall be available to the Inspector for each lot of fittings. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.1.1 The Government Inspector may require magnetic inspection or other Inspection tests whenever Government service experience warrants.

4.2 Examination of Product.- All fittings shall be carefully examined to determine conformance with the requirements for workmanship, finish, marking, and surface defects.

4.3 Sampling.- A maximum of one percent but not less than ten of the fittings from each lot on a contract shall be selected at random for the Inspection tests specified herein. When the sample fittings are not damaged by the Inspection tests, they may be included in the contract or order, except that copper-base alloy fittings that have been subjected to internal strain tests shall be discarded. Rejected and discarded fittings shall not be at the extra cost of the Government.

4.3.1 Lot.- A lot shall consist of all the fittings of a given part number, made from the same batch of material and submitted for inspection at the same time.

4.4 Tests.-

4.4.1 Weight and Dimensions.- Conformance of the fittings with the weight and dimensional requirements of the applicable AN drawings shall be determined by any suitable methods.

4.4.2 Tensile Strength of Steel Fittings.- The tensile strength of steel fittings shall be determined by hardness tests of the fittings, unless otherwise specified on the applicable AN Detail Drawings. Rockwell hardness readings shall be taken on a smooth flat surface of any unthreaded portion of the fitting from which the plating has been removed. Hardness-tensile strength relationships are given in Specification QQ-M-151.

4.4.3 Internal Strain of Aluminum-Bronze and Copper-Base Alloys.- Fittings made of aluminum-bronze or copper-base alloys, as specified herein, shall be immersed in an aqueous solution containing 100 g of mercurous nitrate and 13 ml nitric acid (sp gr 1.43) per liter. After 15 minutes the specimen shall be removed and examined for cracks. Any evidence of cracks indicates internal strain and shall be cause for rejection.

4.4.3.1 Flattening of Sleeves.- Aluminum-bronze sleeves conforming to AN819, which have been relieved as specified herein, shall be squeezed between the smooth jaws of a vise until the shortest distance across the bore is reduced to 40 percent of the inside diameter specified for the sleeve. A satisfactory method of determining the required flattening is to put a stop in the vise jaws equal to dimension T as obtained by the following formula:

$$T = G - 0.60B$$

Where

T is thickness of stop
G is the outside diameter of the sleeve as specified on AN819
B is the diameter of the bore through the sleeve as specified on AN819

This test may be accomplished prior to or after cadmium-plating. Any evidence of cracking of the sleeves shall be cause for rejection.

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4.4.4 Performance Tests for Alternate Materials.- In the event it becomes necessary to substitute materials in the fabrication of the fittings and if these fittings are applicable for use at the operating pressures of hydraulic systems covered by Specification MIL-H-5440, the following Performance tests shall be conducted on five of each size fitting made of such materials. The manufacturer should submit a report of the Performance test with a report on the chemical and physical properties of the material when a request is made for approval of the use of non-standard material. The Procuring Service may require any other tests considered necessary.

4.4.4.1 The Performance test shall be conducted at room temperature and at an ambient air temperature of -54 to -60°C (-65° to -76°F) unless otherwise noted herein. Fittings tested at -54°C (-65°F) should be subjected to this temperature for at least 24 hours at a static pressure of 2 to 10 psi prior to start of the actual test.

4.4.4.2 Assembly and Test of Flared Tube End Fitting.- Each size of fitting for which approval is desired shall be assembled with an approved fitting made of a similar type of material: For example, an aluminum-alloy fitting under test should be assembled with an 52S0 aluminum-alloy tubing, an aluminum-bronze or aluminum-alloy sleeve and an alloy nut. The following procedure shall be followed:

4.4.4.2.1 Pull Up to Minimum Torque.- The fitting assemblies shall be tightened to the minimum wrench torque specified on Drawing AND10064 for at least 15 successive times, each tightening assembly constituting a complete assembly and disassembly of the fitting assembly. After each tightening operation, the fitting assemblies shall be subjected to fluid pressure of 3,000 for 5 minutes. Under this pressure there shall be no leakage or distortion of the assembly. Upon completion of the fifteenth tightening operation, the assemblies shall be subjected to a burst pressure of 6,000 psi for 5 minutes. The fitting assemblies shall not rupture under this pressure.

4.4.4.2.2 Pull Up to Overtightened Torque.- The test specified above for pull up to minimum torque shall be repeated again except that the fitting assemblies shall be tightened to 2-1/2 times the maximum wrench torque specified on Drawing AND10064.

4.4.4.3 Assembly and Test of Pipe-Thread Fittings.- Each size of pipe-threaded fitting submitted for approval shall be assembled on the pipe threads with an approved fitting made of similar type of material. The fittings shall be tightened to the point where leakage shall not occur under a pressure of 3,000 psi plus one complete turn. The tightening shall be accomplished at least 15 successive times, each operation consisting of a complete assembly and disassembly of the fitting from its mating part. After each tightening operation, the fitting assemblies shall be subjected to a pressure of 3,000 psi for 5 minutes. Under this pressure there shall be no leakage or distortion of the fitting assembly. Upon completion of the fifteenth tightening operation the assemblies shall be subjected to, and shall withstand without rupture, a burst pressure of 6,000 psi.

4.4.4.4 Impulse - At Room Temperatures.- Flared tube type fitting assemblies tightened to the minimum wrench torque and to 2-1/2 times the maximum wrench torque specified on Drawing AND10064 shall be subjected to pressure impulse from 0 to 1,500 psi and back to 0 psi at room temperature. The assemblies shall withstand 100,000 complete cyles of impulse at a rate of 35 ±5 cycles per minute without leakage or other malfunctioning. The same test shall be conducted with pipe-threaded fittings which have been tightened sufficiently to prevent leakage plus one additional turn.

4.4.4.5 Assembly and Test of Beaded Type Fittings.- Fittings with a beaded end, for hose connection, if also having a pipe-threaded or flared tube connection on another end shall be subjected to the applicable test as specified above on that connection except that all pressure tests shall be at 500 psi. Hose type connections shall show no leakage when subjected to this pressure.

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4.5 Rejection and Retest.- If 10 percent of the sample fittings fail to conform to any of the requirements of this specification the whole lot shall be tested to those requirements in which the 10 percent of sample fittings failed, except that if 10 percent of the fittings subjected to test for Internal Strain of Copper-Base Alloys fail in such test, the entire lot shall be rejected. Those fittings subjected to other individual tests and failing to meet the requirements shall be rejected. Fittings which have been rejected may be reworked to correct the defects and resubmitted for acceptance, except that copper-base alloy fittings that have been subjected to Internal Strain test shall be discarded whether they pass the test or not. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Procuring Service.

5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by, or direct shipments to the Government.

5.2 Preservation and Packaging.- Fittings in unit quantities specified by the Procuring Service shall be protected from corrosion in accordance with method IA of Specification AN-P-13 except that no preservative compound shall be applied to the fitting. Containers conforming to Specification JAN-P-120, JAN-P-133 or JAN-P-108 shall be used in the preservation process.

5.3 Packing.- Unless otherwise specified, all items shall receive domestic packing. Shipping containers, shall contain the same number of unit packages, shall be uniform in size, and snugly packed. The gross weight of the fully packed container shall not exceed approximately 200 pounds.

5.3.1 Domestic Packing.- Unless otherwise specified, interior packages shall be overpacked in substantial commercial exterior containers so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate,to the point of delivery. Except as specified herein, the container shall conform to the requirements of Consolidated Freight Classification Rules in effect at the time of shipment. The use of corrugated or solid fiberboard having a minimum Mullen test of less than 275 pounds is prohibited. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.3.2 Overseas Packing.- Unless otherwise specified, for overseas shipment, the interior containers shall be packed in boxes, in accordance with Specification JAN-P-105 or JAN-P-106. Plywood, if used, shall be in accordance with Specification JAN-P-139 type B, condition I. A sealed case liner conforming to Specification JAN-P-125 is required.

5.4 Marking and Labeling.- Each interior package shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the packages are opened:

FITTINGS (see applicable AN drawing for title)
AN Part No.
Specification MIL-F-5509
Stock No. (USAF or Navy, as applicable)
Manufacturer's Drawing No.
Name of Manufacturer
Name of Contractor (if different from manufacturer)
Contract or Order No.
Quantity

5.4.2 Shipping Containers.- Each shipping container shall be marked in accordance with the requirements applicable to the individual Services, as specified in Specification 94-40645 for the Air Force, and the Navy Shipment Marking Handbook for the Navy.

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6. NOTES

6.1 Intended Use.- Fittings covered by this specification are intended for use in aircraft.

6.2 Ordering Data.- Requisitions, contracts, and orders should state the AN part number to be furnished, and whether overseas packing is desired.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Other interest:
Air Force

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